Additional file 1: Supplementary tables and figures for "HANFIS: A Hierarchical Adaptive Neuro-Fuzzy Inference System for Evidence-Based Dynamic Cybersecurity Risk Scoring in Cloud-Native Kubernetes Deployments"

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Abstract

Supplementary tables and figures moved out of the main text to shorten it. Each item keeps its original caption and values.

Supplementary tables

Supplementary figures

Table S1 Experimental preparation, acceptance criteria, and methodological purpose.

Preparation element	**Acceptance criterion**	**Methodological purpose**	**Evidence (Runs 7–16)**
Cluster health	Two Kubernetes nodes in Ready state	Excludes measurements obtained during node degradation or loss of schedulability	Gate passed in all ten pairs
Baseline workload P0	Two of two replicas Ready	Holds application capacity constant across monitoring conditions	Gate passed in all ten pairs
Service health	Internal HTTP response code 200	Confirms that the measurement target remains operational before and after state transitions	Gate passed in all ten pairs
Monitoring ON state	Five of five pods Running and fully Ready; Prometheus 1/1 Ready	Defines a complete observability treatment rather than a partially deployed stack	Gate passed in all ten pairs
Monitoring OFF state	Zero monitoring pods remaining	Defines a valid control condition without residual collector workloads	Gate passed in all ten pairs
Protocol freeze	Scripts, controller definitions, services, system fingerprint, and replication plan present	Prevents unrecorded procedural drift between runs	Applied to Runs 7–16; two aborted attempts logged as incidents
Integrity control	SHA-256 manifests validate without mismatch	Detects post hoc alteration or corruption of experimental artefacts	SHA-256 manifest retained for each of the ten run archives
Counterbalancing	Prespecified ON–OFF or OFF–ON order	Reduces order, warm-up, cache, and time-dependent bias	Five ON–OFF and five OFF–ON pairs

Table S2 E1 ablation: membership family × monotonicity, 5 seeds per cell.

Scenario	MF family	Monotone	Params.	RMSE (mean ± SD)	R^2 (mean)
S1 (smooth)	Gauss	no	176	0.093 ± 0.021	1.000
S1 (smooth)	Gauss	yes	176	0.411 ± 0.246	0.999
S1 (smooth)	Ruspini	no	147	0.043 ± 0.052	1.000
S1 (smooth)	Ruspini	yes	147	0.073 ± 0.066	1.000
S2 (threshold)	Gauss	no	176	0.117 ± 0.034	1.000
S2 (threshold)	Gauss	yes	176	6.067 ± 1.524	0.879
S2 (threshold)	Ruspini	no	147	0.646 ± 0.051	0.999
S2 (threshold)	Ruspini	yes	147	2.126 ± 0.906	0.983

Table S3 Post hoc sensitivity of the confirmatory comparison to the EPSS model change: full run versus weeks whose labels all pre-date 2026-06-15 (skill score, 95% group-clustered bootstrap CI).

Hypothesis	Comparator	Full run	Pre-change weeks
H1a	Single CVSS score	0.967 [0.951, 0.980]	0.973 [0.939, 0.991]
H1b	Persistence (4-week)	0.015 [−0.607, 0.246]	−0.055 [−2.731, 0.201]
H1c	Ridge, 7 inputs	0.696 [0.527, 0.816]	0.728 [0.484, 0.901]
H2	FlatANFISMatched	0.665 [0.269, 0.844]	0.913 [0.753, 0.974]
H2′	Flat ANFIS (full)	0.253	0.441

Table S4 Explanation-fidelity checks on the 25 saved models. Pairs are the positive-control images P0-IF with and without the active filter; S^+ denotes the 23 weeks in which exploitability differs between the two.

Check	Criterion	Result	Outcome
H6-0 trace vs. inference	$\lvert\Delta\rvert \leq 10^{-9}$, all test rows	0	pass
H6-neg identical inputs	$\lvert\Delta\rvert \leq 10^{-9}$, 34 weeks × 25 models	0	pass
H6a direction (monotone)	$\hat{y}_{\text{on}} - \hat{y}_{\text{off}} \geq -10^{-9}$ on S^+	minimum 0	pass
H6b panel structure	panel E unchanged, panel V changes	0 and 2.9×10^{-4}	pass
H6c deletion test	ratio ≥ 2 and proportion ≥ 0.90	4.11 and 0.835	fail

Table S5 Weekly ranking utility (E15, post hoc): mean over weeks of the Spearman correlation between predicted score and realised label, and precision at the top 20%.

Horizon and metric	Formula	HANFIS	Persistence
4 weeks (45 weeks): Spearman	0.994	0.969	0.992
4 weeks: precision at top 20%	0.969	0.973	0.969
8 weeks (37 weeks): Spearman	0.989	0.963	0.982
8 weeks: precision at top 20%	0.957	0.957	0.941
12 weeks (29 weeks): Spearman	0.974	0.951	0.964
12 weeks: precision at top 20%	0.924	0.952	0.883

Table S6 Paired weekly differences in ranking utility (E15, post hoc); 95% week-bootstrap intervals in brackets.

Comparison	Δ Spearman	Δ precision at top 20%
4 w: formula − persistence	+0.002 [0.001, 0.004]	0.000 [−0.013, 0.013]
4 w: HANFIS − persistence	−0.023 [−0.027, −0.018]	+0.004 [−0.009, 0.018]
4 w: formula − HANFIS	+0.025 [0.021, 0.029]	−0.004 [−0.013, 0.000]
8 w: formula − persistence	+0.007 [0.004, 0.012]	+0.016 [0.000, 0.032]
8 w: HANFIS − persistence	−0.018 [−0.026, −0.011]	+0.016 [0.000, 0.032]
8 w: formula − HANFIS	+0.026 [0.020, 0.031]	0.000 [0.000, 0.000]
12 w: formula − persistence	+0.010 [0.005, 0.017]	+0.041 [0.014, 0.076]
12 w: HANFIS − persistence	−0.013 [−0.025, −0.002]	+0.069 [0.034, 0.103]
12 w: formula − HANFIS	+0.024 [0.017, 0.032]	−0.028 [−0.055, −0.007]

Table S7 Input-change strata at four weeks (E16, post hoc): rows, mean squared error (persistence / formula / HANFIS), and skill against persistence with 95% intervals.

Stratum	Rows	MSE	Skill, formula	Skill, HANFIS
Input changed	628	0.637 / 0.382 / 0.557	0.400 [0.308, 0.487]	0.125 [−0.402, 0.333]
No input change	452	0.107 / 0.106 / 0.202	0.002 [−0.069, 0.037]	−0.900 [−4.171, −0.206]

Table S8 Calibration of four-week out-of-fold predictions (E21, post hoc). Mean error (prediction minus label), and slope β and intercept α of the regression of the label on the prediction, with 95% image-bootstrap intervals. Unaffected rows are those whose label window does not straddle the change of the EPSS model.

Predictor	Mean error	Slope β	Intercept α
All rows (1080)			
Persistence	−0.097 [−0.196, −0.013]	0.997 [0.986, 1.007]	0.109 [0.012, 0.228]
Formula	−0.040 [−0.106, 0.017]	0.998 [0.990, 1.005]	0.048 [−0.021, 0.132]
HANFIS	0.016 [−0.080, 0.104]	1.025 [1.008, 1.044]	−0.115 [−0.228, 0.007]
Unaffected rows (890)			
Persistence	−0.015 [−0.079, 0.025]	0.997 [0.993, 1.001]	0.026 [−0.017, 0.091]
Formula	0.012 [−0.029, 0.040]	1.000 [0.997, 1.002]	−0.012 [−0.044, 0.035]
HANFIS	0.068 [−0.029, 0.157]	1.031 [1.014, 1.053]	−0.193 [−0.300, −0.085]



Fig. S1 Methodological mind map linking the cloud-native testbed, telemetry acquisition, monitoring-overhead controls, reproducibility evidence, and HANFIS integration.

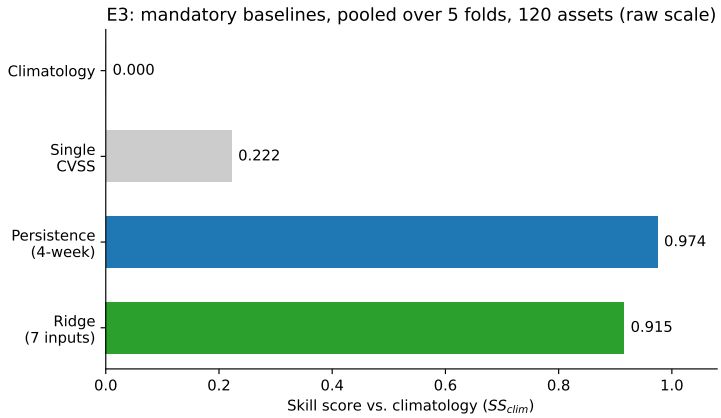


Fig. S2 Mandatory baselines' skill against climatology, pooled over 5 folds. A single CVSS score now adds modest skill over climatology (22%, up from essentially nothing on the earlier 27-image pilot); persistence and ridge separately recover over 90% of the climatology-relative skill each, though not of each other; see the mandatory-baselines table in the main text for the reversal in their relative standing.

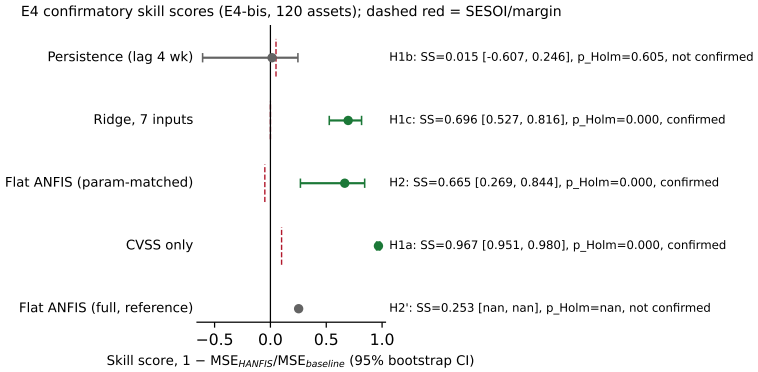


Fig. S3 E4 confirmatory results: HANFIS skill score against each comparator with 95% group-clustered bootstrap confidence intervals. Green bars exclude zero (confirmed); the red bar (H1b, vs. persistence) does not.